Benchmarking swarm coordination algorithms for factory drone patrol

Market-based auction, artificial potential fields, a PPO leg planner and NMOPSO in PX4/Gazebo: methods, metrics, results and recommendations

Benchmark suite `suite_final` · 68 simulated missions · generated 2026-10-23

Abstract

Indoor factory patrol with small multirotors needs coordination that is efficient, safe in cluttered aisles and robust to the radio fading typical of metal-rich halls. We benchmark four coordination approaches on one platform (ROS 2 Jazzy, PX4 SITL, Gazebo Harmonic, three x500 quadrotors): a centralised sequential single-item **auction** with fixed transit layers, reactive **artificial potential field (APF)** navigation, a **reinforcement-learning (PPO) leg planner** trained on the NMOPSO path model, and **NMOPSO** itself, a multi-objective particle swarm path planner. All four share the same task allocation, failure detector and measurement pipeline, so differences are attributable to navigation and fault behaviour. Four scenarios (nominal patrol, a spatial-conflict stress test, and two communication-blackout faults) are repeated and scored on efficiency, effectiveness, safety, fairness, coordination overhead and robustness metrics, following a recent review framework for industrial drone swarms. No approach dominates. Auction and APF finish the nominal patrol fastest (44.1 ± 0.3 s and 44.3 ± 0.2 s, against 58.7 ± 0.2 s for RL and 56.1 ± 0.3 s for NMOPSO), and APF flies the shortest paths. In the cluttered stress scenario, however, APF crashed a drone in 5 of 5 runs and the auction in 3 of 5, while the two planners completed every run (RL 5, NMOPSO 2) with no obstacle or drone contact. NMOPSO, with its legs planned ahead and cached, matches RL's safety without RL's planning stalls (worst decision 1.9 ± 2.3 ms against 1600 ± 93 ms). Under communication loss the auction recovers fastest because its drones patrol autonomously without radio, while APF, RL and NMOPSO drones depend on ground-station commands and hover. For a factory, where a collision is unacceptable, a planning-based navigator is the safest choice, provided it is combined with on-board fallback behaviour and a replicated allocator.

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1 Introduction

Drone swarms are increasingly proposed for routine inspection in manufacturing plants: reading gauges, checking machines and racks, and watching for anomalies between scheduled maintenance rounds. A factory hall is a difficult airspace. It is GPS-denied, dense with tall racks and machines, shared with people, and its steel structure causes multipath fading that can silence a drone's radio for seconds at a time. A coordination algorithm for such a swarm must therefore be judged not only by how fast it finishes a patrol but also by how close it lets drones come to obstacles and to each other, how evenly it spreads work, how much coordination it costs, and how it behaves when a drone drops off the network.

This report benchmarks four coordination approaches that represent three families in the literature, implemented on a common simulation platform:

- **Auction** (market-based): a centralised sequential single-item auction allocates inspection tasks; drones fly between tasks on fixed per-drone altitude layers through a shared centre corridor.
- **APF** (behaviour-based): the same allocation, with reactive navigation by artificial potential fields (goal attraction, Khatib repulsion from obstacles and from other drones).
- **RL (vB)** (learning-based): the same allocation, with each leg planned by a proximal policy optimisation (PPO) policy trained on the NMOPSO multi-objective path model, conditioned on a threat budget and on a precomputed shortest-route hint.
- **NMOPSO** (swarm-intelligence planner): the same allocation, with each leg planned by navigation-variable multi-objective particle swarm optimisation (Duong, Bui and Phung, 2025), the optimiser whose problem the RL policy was trained on.

The comparison is deliberately controlled: all four use the same on-board auction for task allocation, the same heartbeat-based failure detector, the same PX4 flight stack and the same instrumentation. Differences in the results therefore come from *navigation* (how a drone gets from one task to the next) and from *what a drone does when it has no radio*. The metric set follows the ten core metrics of a recent review framework for industrial drone networks (mission completion time, task completion rate, flight distance, energy, workload imbalance, allocation time, task reallocations, messages exchanged, minimum inter-drone distance and failure recovery time), extended with obstacle clearance, patrol idleness and decision latency.

Section 2 describes the platform and scenarios, Section 3 the algorithms with their equations, Section 4 every metric and the statistical method, Section 5 the results metric by metric, Section 6 discusses suitability for factory deployment, Section 7 concludes and Section 8 lists limitations.

2 Simulation platform and scenarios

2.1 Architecture

Every experiment runs the full flight stack in software-in-the-loop (Figure 1). Gazebo Harmonic simulates a 30 m × 20 m × 12 m factory hall with 18 obstacle boxes (rack rows, machines, a partition, conveyors). Three PX4 SITL instances fly x500 quadrotors in offboard position control and expose their uORB topics to ROS 2 through a Micro XRCE-DDS agent. On every drone a C controller (rclc) runs the patrol state machine, the auction (on drone 0), the heartbeat monitor and the fault injector. A ground-station coordinator (Python) publishes the mission and, for APF, RL and NMOPSO, runs the navigation algorithm at 10 Hz and publishes one MovementCommand per drone. A metrics collector records ground-truth states and events and computes every metric.

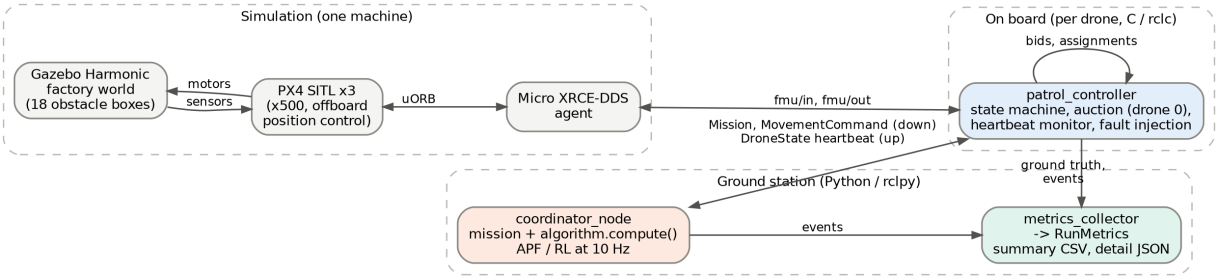


Figure 1. Platform architecture. Radio traffic (DroneState heartbeats, bids, assignments, movement commands) is what the fault injector silences; the ground-truth state and events are instrumentation and are never silenced.

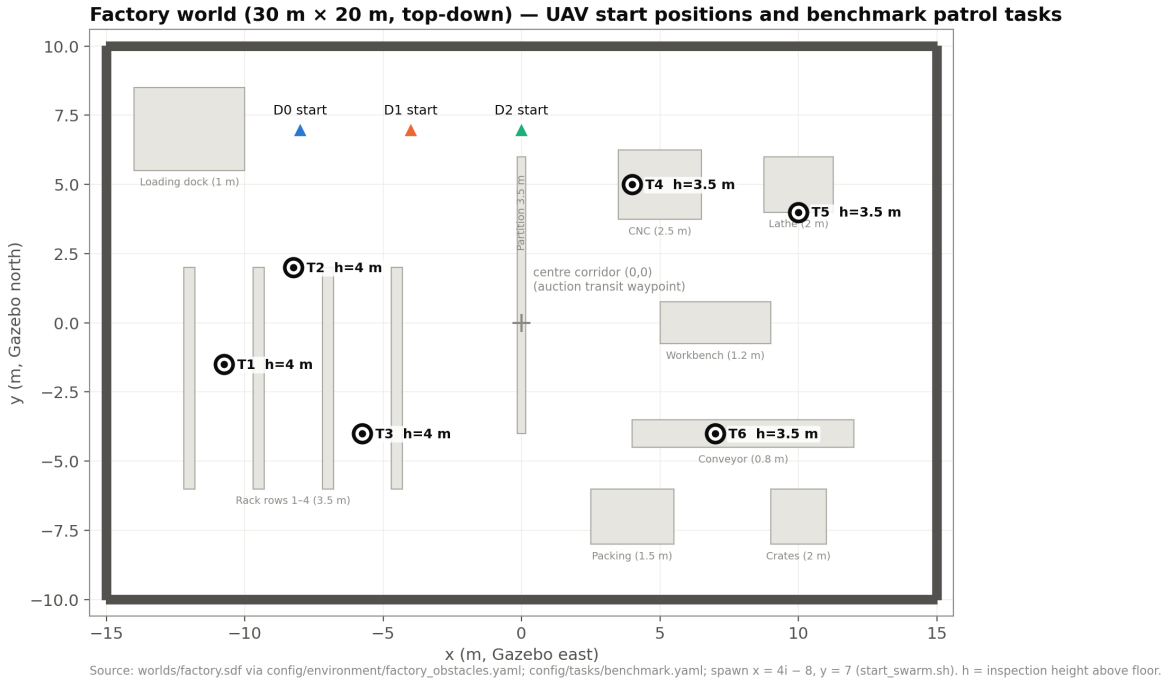


Figure 2. Factory world (top view) with the drone start positions and the six benchmark tasks (T1–T6 = task ids 101–106). The cross at (0, 0) is the auction's transit corridor.

2.2 Scenarios and protocol

Table 1. Scenarios. A blackout silences the drone's radio both ways (no heartbeat, bids, assignments or commands), while it keeps flying its own state machine.

Scenario	Tasks	Fault	Window	Purpose
benchmark	6 tasks over racks and machines (Figure 2), 4 s inspection hover	none	until every task is inspected once	nominal efficiency
safety_stress	8 tasks: three collinear rack-aisle tasks under transit routes, a task beside the centre corridor (1.5, 0.5, 4.5 m), a low task inside a rack aisle (2.0 m)	none	until every task is inspected once (watchdog 330 s)	spatial conflicts, safety
comm_loss	benchmark tasks	drone 1: radio silent from t + 20 s for 15 s	150 s	failure detection and recovery
comm_loss_orchestrator	benchmark tasks	drone 0 (auctioneer and heartbeat monitor) silent from t + 20 s for 15 s	150 s	single point of failure

Each of the 12 algorithm–scenario cells is repeated five times; runs are interleaved (repeat 1 of every cell before repeat 2) so slow drift of the host affects all algorithms alike, and every run is started from a clean process state by a watchdog that also records why a run stopped. Runs stopped for an infrastructure reason (disk, log size) are excluded; runs in which a drone crashes are kept and counted, because a crash is an outcome of the algorithm.

Two properties of the platform shape the interpretation and are stated up front. First, the simulated battery is a time counter (0.1 % per second of mission per drone), so battery consumption measures flight time, not physical energy. Second, in every algorithm a drone inspects its first assigned task twice in a row (an artefact of the controller's zone loop that affects all four equally), which lengthens every mission by roughly one dwell.

3 Coordination algorithms

A coordination algorithm answers two questions: *who* does which task (allocation) and *how* each drone gets there without collisions (navigation). In this benchmark all four algorithms share the allocation layer of Section 3.1; they differ in navigation (Sections 3.2–3.5).

3.1 Shared allocation: centralised sequential single-item auction

When the mission arrives, all M tasks are unassigned. After 5 s (so that every peer state has arrived), drone 0 acts as auctioneer and assigns the tasks one at a time, in file order. The cost of drone i for task j combines the 3-D distance to the task, the drone's current load and its battery deficit:

$$c_{ij} \;=\; |\mathbf{p}_i - \mathbf{q}_j|_2 \;+\; 15\, n_i \;+\; 0.2\,(100 - b_i) \tag{1}$$

where p_i is the drone position, q_j the task position, n_i the number of tasks already assigned to drone i and b_i its battery level in percent. Task j goes to the cheapest eligible drone; eligibility caps the load so that every healthy drone receives an equal share:

$$i^*(j) \;=\; \arg\min_{i \,\in\, \mathcal{E}} c_{ij}\,, \qquad \mathcal{E} \;=\; \{\, i \;:\; h_i = 1,\;\; n_i < \lceil M/N_h \rceil \,\} \tag{2}$$

with h_i the health flag and N_h the number of healthy drones. Each drone then loops over its own tasks indefinitely (continuous patrol). Every controller also publishes bids, but the auctioneer recomputes the costs from its own view of the peer states, so the mechanism is a centralised greedy auction run on board drone 0, not a distributed consensus protocol.

Failure detection and reallocation. Every drone broadcasts its state at 10 Hz; this is the heartbeat. Drone 0 declares a peer FAILED when it has been silent for longer than the timeout, or when its battery is below 20 %, revokes its tasks and re-auctions them among the remaining drones:

$$t - t_i^{\text{last}} > T_{hb} = 5\ \text{s} \ \Rightarrow\ h_i \leftarrow 0,\ \ \{j : i^*(j) = i\} \rightarrow \text{re-auction} \tag{3}$$

A drone whose heartbeat returns is marked healthy again and can win tasks in later auctions. The scheme is simple and fast, but drone 0 is a single point of failure: if drone 0 itself goes silent, no other drone runs the detector or the auction (scenario comm_loss_orchestrator).

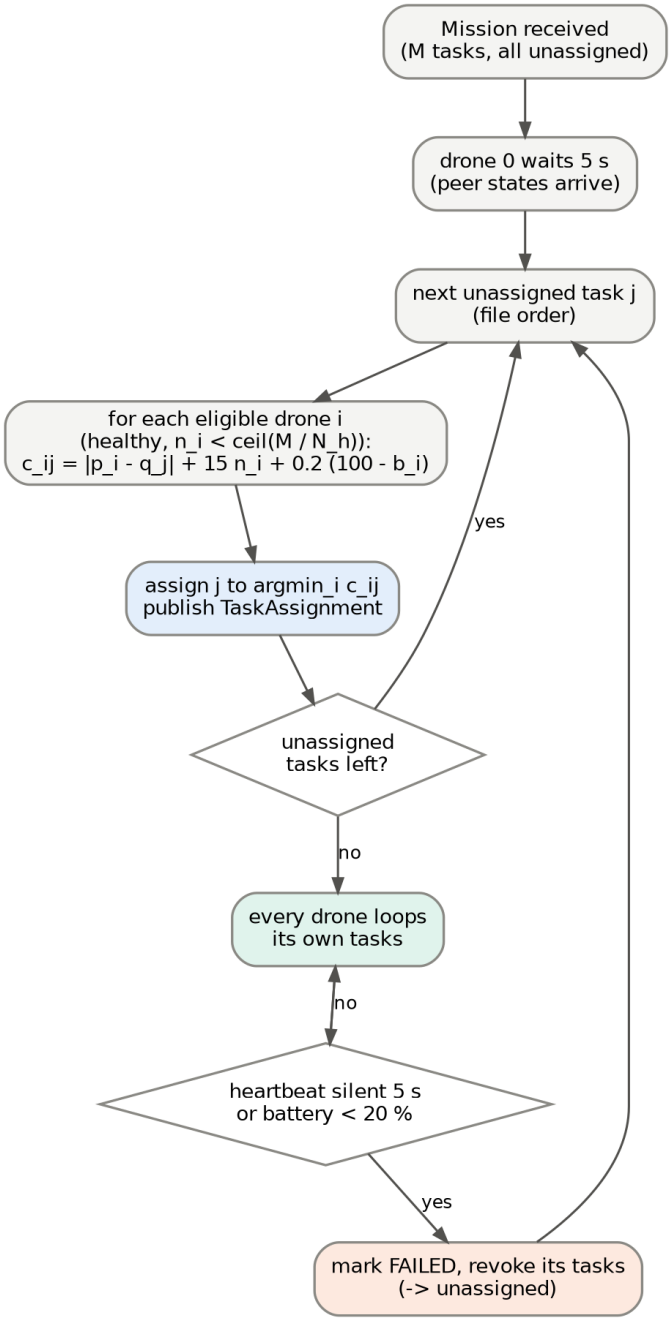


Figure 3. Allocation and failure handling (shared by all four algorithms).

3.2 Auction navigation: altitude layers and a centre corridor

With the auction algorithm, each drone also navigates on board (Figure 4). Drones take off staggered by 3 s per id, climb to a private transit layer, fly to the centre corridor (0, 0), then to above the task, and descend to the inspection point:

$$h_i^{\text{transit}} = 5.0 + 1.25\,i \ \text{ m} \ \ (\le 10 \text{ m}), \qquad h_i^{\text{patrol}} = 2.5 + i \ \text{ m} \tag{4}$$

Separate layers keep drones apart vertically, and the corridor keeps transits away from walls. A proximity rule resolves remaining conflicts: a drone holds while a lower-id drone is within 2.5 m of its target. The method needs no obstacle model; it is safe as long as every obstacle is below the lowest layer and the corridor is free, and it relies on peer positions arriving over the radio.

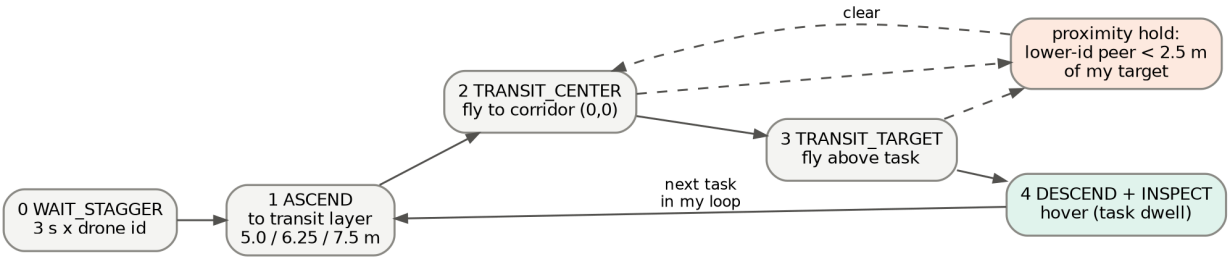


Figure 4. On-board navigation state machine of the auction algorithm. APF, RL and NMOPSO replace states 1–3 by external movement commands; state 4 (descend and inspect) is shared.

3.3 APF: artificial potential field navigation

APF is a reactive, behaviour-based method: at every tick (10 Hz) the ground station sums three virtual forces on each active drone and turns the result into a velocity command. The attraction pulls the drone towards its goal g with a saturated linear law:

$$\mathbf{F}_{att} = k_{att}\,(\mathbf{g} - \mathbf{p}), \qquad |\mathbf{F}_{att}| \le v_{max} \tag{5}$$

Obstacles and other drones repel with the Khatib potential: for every obstacle box k with surface distance d_k inside the influence radius ρ_o, and every other drone m within ρ_d,

$$\mathbf{F}_{obs} = \sum_{k\,:\,d_k < \rho_o} k_o\left(\tfrac{1}{d_k} - \tfrac{1}{\rho_o}\right)\tfrac{1}{d_k^2}\,\mathbf{n}_k, \qquad \mathbf{F}_{drone} = \sum_{m\,:\,d_m < \rho_d} k_d\left(\tfrac{1}{d_m} - \tfrac{1}{\rho_d}\right)\tfrac{1}{d_m^2}\,\mathbf{n}_m \tag{6}$$

where n is the unit vector pointing away from the obstacle or drone and d is floored at 0.1 m. The command is smoothed, clipped and sent as a position set-point one horizon ahead:

$$\mathbf{v}_t = \text{clip}_{v_{max}}(0.8\,(\mathbf{F}_{att} + \mathbf{F}_{obs} + \mathbf{F}_{drone}) + 0.2\,\mathbf{v}_{t-1}), \qquad \mathbf{p}^{sp} = \mathbf{p} + \tau\,\mathbf{v}_t \tag{7}$$

Table 2. APF parameters (config/algorithms/apf.yaml); identical in every scenario, not tuned per scenario.

Parameter	Symbol	Value	Meaning
goal gain	k_att	0.5 1/s	attraction slope
maximum speed	v_max	1.5 m/s	limit on attraction and command
obstacle gain / radius	k_o / ρ_o	0.5 / 1.5 m	repulsion balances full attraction at ~0.6 m
drone gain / radius	k_d / ρ_d	10 / 3.0 m	repulsion balances full attraction at ~1.5 m
goal tolerance	–	1.2 m	inside it, the command is the goal itself
smoothing / horizon	–, τ	0.2 / 1.0 s	velocity filter, set-point look-ahead

APF needs an obstacle model but no planning: decisions take well under a millisecond. Its known weaknesses are local minima (forces cancel before the goal) and the trade-off between repulsion strength and the ability to reach goals close to obstacles: inspection points sit next to racks and machines, so the repulsion cannot be made arbitrarily strong.

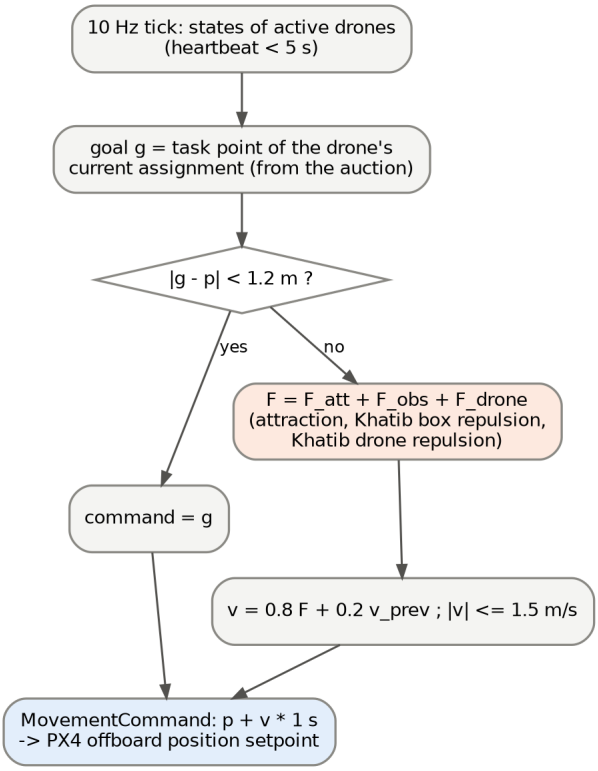


Figure 5. APF control loop (ground station, 10 Hz).

3.4 RL (vB): a PPO leg planner on the NMOPSO path model

The learning-based approach replaces navigation by a policy that plans each leg (the path from the drone's position to its next task). It was trained with proximal policy optimisation (PPO) on the leg-planning problem of NMOPSO, a multi-objective particle-swarm path planner, so that its outputs are ordinary NMOPSO paths and can be scored with NMOPSO's own cost.

Path model and action

A path has n = 10 nodes between start s and goal g. Node k is encoded in spherical coordinates (r_k, ψ_k, φ_k) relative to the previous node, with chained transforms:

$$\mathbf{T}_k = \mathbf{T}_{k-1}\, R_z(\phi_k)\, R_y(-\psi_k)\, T_x(r_k), \qquad r_k \in [\tfrac{r_{max}}{9},\, r_{max}],\;\; r_{max} = \tfrac{3\,|\mathbf{g}-\mathbf{s}|}{10},\;\; |\psi_k|, |\phi_k| \le \tfrac{\pi}{4} \tag{8}$$

An episode has one step per node. The action $a \in (-1, 1)^3$ is a residual around the straight path: a_r scales the even-spacing step r_{nom} (remaining distance over remaining segments) by e^{a_r}; a_ψ turns the bearing from the goal bearing by up to ±90°; a_z sets the height inside the patrol band. The zero action is "straight at the goal".

Objectives (NMOPSO MyCost)

A completed path P is scored with four objectives: length overhead, threat (proximity to obstacles modelled as cylinders of radius R_m), altitude deviation and turning:

$$J_1 = \left|1 - \tfrac{|\mathbf{g}-\mathbf{s}|}{L(P)}\right|, \qquad J_2 = \tfrac{1}{SM}\sum_{s=1}^{S}\sum_{m=1}^{M} \mathrm{clip}_{[0,1]}\!\left(1 - \tfrac{d_{sm} - \delta - R_m}{10\,\delta}\right) \tag{9}$$

$$J_3 = \tfrac{1}{n}\sum_k \tfrac{|z_k - \bar{z}|}{\Delta z/2}, \qquad J_4 = \tfrac{1}{n}\sum_k \tfrac{\angle(\mathbf{e}_k, \mathbf{e}_{k+1})}{\pi} \tag{10}$$

Here d_{sm} is the distance from segment s to cylinder m, δ the drone size (0.1 m), L(P) the path length, $\bar{z}$ and Δz the centre and height of the band, and e_k the segment directions. A path is infeasible (J_1 or $J_2 = \infty$) if a segment is too short or enters an inflated cylinder.

Reward, threat budget and conditioning

Each training episode draws a preference w over (J_1, J_3, J_4) and a threat budget u ∈ [0.15, 1.2] that limits J_2 to u times the threat of the straight path; both are observed by the policy. The terminal reward of a feasible path is the negative scaled cost plus a penalty for exceeding the budget, capped at S_{max} = 5; an infeasible path receives a lower reward than any feasible one:

$$r_T = -\min\left(\sum_{i\neq 2}\frac{w_i J_i}{F_i} + \lambda_b \frac{\max(0,\, J_2 - u J_2^{str})}{F_2},\ S_{max}\right), \qquad r_T^{infeasible} = -(S_{max} + 1 + \min(v, 3)) \tag{11}$$

with F_i fixed scales (the 95th percentile of NMOPSO fronts), λ_b = 50 and v a soft violation measure. In vB the threat term is paid densely, segment by segment, so that credit reaches the step that caused it (the return is unchanged), and a straight-path prior $-0.02\ \|a_k\|^2$ per step keeps the policy on the straight line unless deviating pays. The observation (40 values) contains the goal in the chain frame, the reach left, height, clamping offset, steps left, leg length, heading relative to the goal bearing, 16 obstacle range rays, a route hint (bearing of the farthest visible waypoint of the shortest collision-free route, route length over the reach left and over the straight distance), the budget left and the straight path's threat-to-go, the budget u and the preference w.

Training

The policy is trained with PPO-clip. With probability ratio $\rho_t(\theta) = \pi_\theta(a_t|o_t) \,/\, \pi_{\theta_{old}}(a_t|o_t)$ and generalised advantage estimates $\hat{A}_t$,

$$L^{CLIP}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t,\ \mathrm{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\right)\right], \qquad \hat{A}_t = \sum_{l\geq 0}(\gamma\lambda)^l \delta_{t+l} \tag{12}$$

Table 3. Training configuration of the released policy vB (seed 2 of three trained seeds).

Setting	Value	Setting	Value
iterations × episodes	2000 × 8192 legs	actor / critic	MLP 3×256 tanh, conditioning fed to every layer
PPO clip ε / KL stop	0.2 / 0.02	learning rate actor / critic	$3\cdot10^{-4}$ / 10^{-3}
γ, λ (GAE)	1.0, 1.0 (10-step episodes)	epochs per iteration	10
policy	tanh-squashed Gaussian	training legs	70 % with real-leg lengths, rest random
budget penalty λ_b	50	straight prior	0.02 ‖a‖²

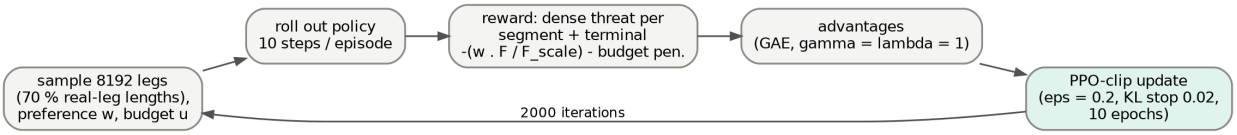


Figure 6. PPO training loop.

Deployment

At run time (flowchart at the end of this section), the planner rolls out the policy for 48 conditions (12 threat budgets × 4 preferences), one deterministic and three sampled rollouts each, plus the straight path: 193 candidates per leg, scored with the same cost. Among the feasible candidates it picks the "short-safe" path:

$$\mathcal{C} = \{\, c : J_2^c \leq \min_{\tilde{c}} J_2^{\tilde{c}} + 0.1\,(\max_{\tilde{c}} J_2^{\tilde{c}} - \min_{\tilde{c}} J_2^{\tilde{c}})\,\}, \qquad c^* = \arg\min_{c\in\mathcal{C}} J_1^c \tag{13}$$

If no candidate is feasible, a cross-entropy refinement is tried, and then a decomposition of the leg along an A* route into line-of-sight sub-legs, each planned by the policy. Each drone flies its path on its own layer (1.8 m + 1.0 m × id) with a carrot follower at 1.5 m/s and yields to a

lower-id drone within 2.5 m. There is no fallback to an unchecked straight line: a leg without a feasible plan is held and retried. The planner runs on the coordinator's single thread, so a slow plan delays the commands of every drone.

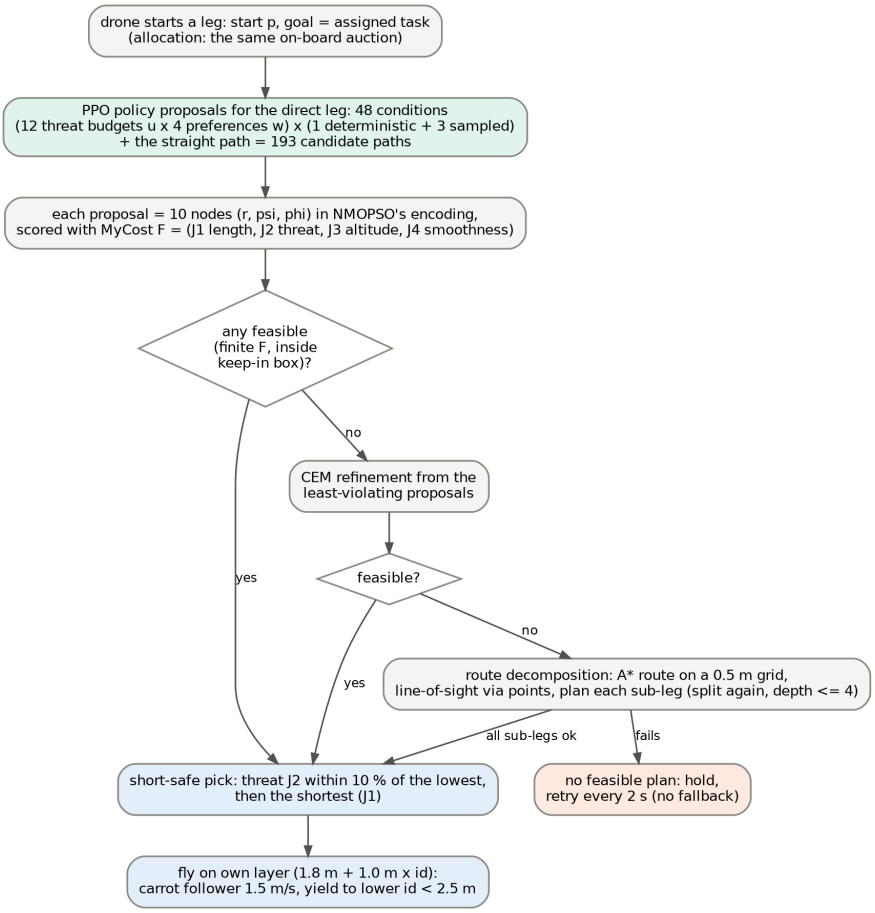


Figure 7. RL leg planning at run time.

3.5 NMOPSO: multi-objective particle swarm leg planning

NMOPSO (navigation-variable-based multi-objective particle swarm optimisation; Duong, Bui and Phung, Neural Computing and Applications 37, 2025) solves the same leg-planning problem as the RL policy: the same 10-node navigation-variable encoding (Eq. 8), bounds and MyCost objectives J_1–J_4 (Eqs. 9–10). Instead of a trained policy it searches each leg with a swarm of P particles $x_i = (r, \psi, \varphi) \in \mathbb{R}^{3n}$, each with velocity v_i and personal best p_i:

$$\mathbf{v}_i \leftarrow w\,\mathbf{v}_i + c_1\,\mathbf{r}_1 \odot (\mathbf{p}_i - \mathbf{x}_i) + c_2\,\mathbf{r}_2 \odot (\mathbf{l}_i - \mathbf{x}_i), \qquad |\mathbf{v}_i| \le \tfrac{1}{2}(x_{max} - x_{min}), \qquad \mathbf{x}_i \leftarrow \mathrm{clip}(\mathbf{x}_i + \mathbf{v}_i) \tag{14}$$

with r_1, r_2 uniform random vectors, the velocity mirrored where a variable left its bounds, and inertia w damped by 0.98 per iteration. Because the four objectives conflict, there is no single global best. NMOPSO keeps a repository R of non-dominated paths (Pareto dominance: a ≺ b if a is no worse in every objective and better in one) and draws each particle's leader l_i from a hypergrid over R, preferring sparsely occupied cells:

$$P(\text{cell } c) = \frac{e^{-\beta N_c}}{\sum_{c'} e^{-\beta N_{c'}}}, \qquad p_m(t) = \left(1 - \frac{t-1}{T-1}\right)^{1/\mu}, \qquad \mathbf{x}_i^{mut} = \mathbf{x}_i + \mathcal{N}(0,1) \odot \mathbf{p}_i \tanh(\delta\,|R|) \tag{15}$$

N_c is the number of repository members in cell c. With the decaying probability p_m a particle is mutated; the mutant replaces it if it dominates it, or with probability 0.5 if neither dominates. When R exceeds its size, members of crowded cells are deleted (probability $\propto e^{\gamma N_c}$). After the last iteration, the deployed path is the short-safe member of R (Eq. 13), the same rule as for

RL.

Table 4. NMOPSO parameters: the authors' published code, unchanged.

Parameter	Value	Parameter	Value
swarm size P	100	iterations T	500
repository size	50	inertia w, damping	1.0, 0.98
c_1, c_2	1.5, 1.5	grid divisions per objective	5 (inflation 0.1)
leader / deletion pressure β, γ	2, 2	mutation μ, δ	0.5, 20

Implementation and planning time

The planner is a vectorised port of the authors' MATLAB NMOPSO.m: the whole swarm is decoded and scored in one batched MyCost evaluation per iteration (PyTorch, verified against the MATLAB cost), which cuts the time per leg from a median of 369 s for the MATLAB original on the 73 factory legs to about 5 s. One detail of the authors' code is corrected: their mutation clamps every mutated variable to its lower bound; the port clamps to [x_{min}, x_{max}] as the paper describes. Five seconds is still fifty coordination ticks, so planning never runs inside the 10 Hz loop:

- **Plan cache.** The legs between the scenario's fixed points (each drone's take-off point to every task, and every task to every other task) are planned before the run, in parallel, with the same planner and seeds; at run time a leg starting within 1 m of a cached start uses the cached path.
- **Background planning.** Any other leg (for example from wherever a drone was holding when its radio returned) is planned in a worker thread while the drone holds its position; the result is applied when it arrives and added to the cache.
- **Shared remainder.** Route decomposition of blocked legs, the per-drone layers, the carrot follower and the yield rule are the RL algorithm's code, so RL and NMOPSO differ only in how the candidate paths of a leg are produced.

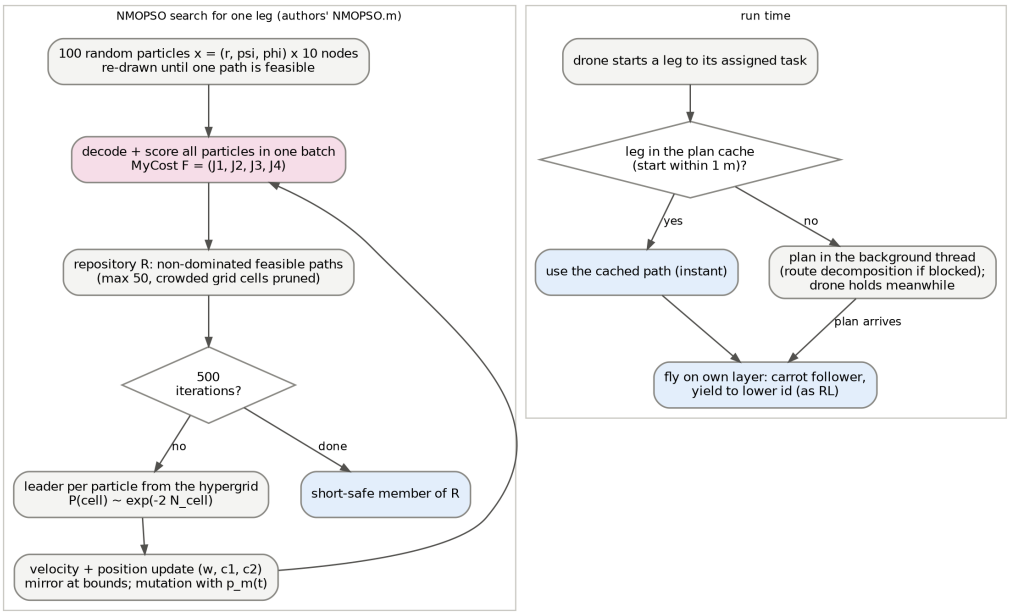


Figure 8. NMOPSO search for one leg (left) and its use at run time (right).

3.6 Summary of the four approaches

Table 5. Properties of the four approaches.

	Auction	APF	RL (vB)	NMOPSO
Family	market-based allocation + rule-based navigation	behaviour-based (reactive)	learning-based planner	swarm-intelligence planner
Allocation	sequential single-item auction (drone 0)	same	same	same
Navigation runs	on board	ground station, 10 Hz	ground station, per leg + 10 Hz follower	cached / background per leg + 10 Hz follower
Obstacle model	none (altitude layers, corridor)	18 boxes	87 cylinders (NMOPSO model)	87 cylinders (NMOPSO model)
Drone deconfliction	altitude layers + proximity hold	repulsion (3 m radius)	altitude layers + yield rule	altitude layers + yield rule
Without radio	keeps patrolling its own tasks	holds its last set-point	holds its last set-point	holds its last set-point
Planning cost	negligible	< 1 ms	10–2000 ms per plan, in the control loop	about 5 s per leg, outside the control loop

4 Metrics and methodology

All metrics are computed by the same collector for every algorithm, from the controllers' ground-truth state stream (10 Hz per drone) and their events (task completed, task assigned, fault and compute-time events). Positions are interpolated onto a common 20 Hz timeline for every distance-based metric. The mission window runs from the mission dispatch to the first completion of the last task; in the fault scenarios an observation window of 150 s is used for fault, safety and patrol metrics, so that recovery after the first patrol cycle is measured.



Figure 9. Measurement pipeline.

4.1 Efficiency and effectiveness

$$\mathrm{MCT} = t_{end} - t_0, \qquad \mathrm{TCR} = 100\,\frac{|\{j : j \text{ completed}\}|}{M}, \qquad \mathrm{TFD} = \sum_i \sum_k |\mathbf{p}_{i,k} - \mathbf{p}_{i,k-1}| \tag{16}$$

MCT is the mission completion time (dispatch to the completion of the last not-yet-completed task), TCR the task completion rate and TFD the total 3-D flight distance over all drones. Battery consumed is reported for completeness; because the simulated battery is a time counter, it is proportional to mission time. Patrol **idleness**, the standard multi-robot patrolling measure, is the time since a task was last inspected, $I_j(t) = t - $ (last visit of j); we report its time average over tasks and the window (average idleness) and its maximum (worst coverage gap):

$$\bar{I} = \frac{1}{MT} \sum_{j=1}^{M} \int_0^T I_j(t)\, dt, \qquad I_{max} = \max_{j,\,t} I_j(t) \tag{17}$$

4.2 Safety

Obstacle clearance is the distance from a drone's centre to the nearest obstacle box (0 inside a box; the floor is excluded and only airborne samples above 0.5 m count). A *near miss* is time below 0.6 m (twice the airframe half-size), a *contact* an episode below 0.3 m, where the airframe touches the box. Between drones, the review framework's **minimum inter-drone**

distance is complemented by the swarm separation profile, its percentiles and violations of the safety radius d_{safe} = 1.0 m:

$$s(t) = \min_{i \neq j} |\mathbf{p}_i(t) - \mathbf{p}_j(t)|, \qquad \text{MIDD} = \min_t s(t), \qquad s_{p1}\,,\ s_{p5} = 1\text{st},\ 5\text{th percentile of } s(t) \tag{18}$$

Violations count episodes with $s(t) < d_{safe}$; drone contacts count episodes of a pair closer than 0.6 m. A crashed drone (outside the hall, on the ground while in a flying state, or reporting FAILED) is excluded from later position samples and its run is counted as a run with a crash.

4.3 Fairness

Workload imbalance is computed on two workload definitions, as the review framework recommends: task completions per drone and flight distance per drone, w_i, over the evaluation window. We report one minus Jain's fairness index and the Gini coefficient:

$$\text{WI} = 1 - \frac{(\sum_i w_i)^2}{N \sum_i w_i^2} \in [0,\ 1 - \tfrac{1}{N}], \qquad G = \frac{\sum_i \sum_j |w_i - w_j|}{2 N^2 \bar{w}} \tag{19}$$

4.4 Coordination overhead

Allocation latency is the time from mission dispatch to the first task assignment and to the moment every task has been assigned once; the ratio of the latter to MCT is the share of the mission spent organising (the framework flags ratios above 0.3 as coordination-bound). **Task reallocations** count assignments that move a task to a different drone (re-assigning a task to the same drone is not counted), in total and per minute. **Decision time** is the wall time of one navigation decision in the coordinator (APF, RL and NMOPSO; the mean, 95th percentile and maximum), and allocation time the on-board auction step. Message volume (MBE in the framework) is not yet instrumented; Section 8 discusses it.

4.5 Robustness to communication loss

For the injected blackout starting at t_0 and ending at t_1, the failure-recovery time of the framework is decomposed into its components:

$$t_{detect} = t_{FAILED} - t_0, \quad t_{reassign} = t_{1st\ reassignment} - t_0, \quad T_{recovery} = \max_{j \in \mathcal{L}} t_{revisit}(j) - t_0 \tag{20}$$

where $\mathcal{L}$ is the set of tasks the lost drone held. Further: *rejoin* (link back to heartbeat accepted), *back in service* (link back to the lost drone's next completed task), the minimum separation between the lost drone and the others while positions are not shared, and the framework's *transient safety degradation* over $[t_0,\ t_1 + 10\ \text{s}]$: its depth, $\max(0,\ d_{safe} - \min s(t))$, and its duration, the time with $s(t) < d_{safe}$.

4.6 Statistics

Each cell has n = 5 runs for auction, APF and RL and n = 2 for NMOPSO (fewer for metrics that only exist in completed runs). We report the mean with a 95 % confidence interval from the t distribution (n − 1 degrees of freedom). An algorithm is called best on a metric only if its interval does not overlap the runner-up's; otherwise the result is a tie. Because several metrics are bounded or skewed, we additionally apply the non-parametric Kruskal–Wallis test across the four algorithms and pairwise two-sided Mann–Whitney U tests with Holm correction, reporting the rank-biserial correlation r as effect size (Appendix C). With five runs per group, the smallest attainable pairwise p-value is 0.008, so only consistent, complete separations reach significance; for NMOPSO (two runs against five) it is 0.095, so no pairwise test involving NMOPSO can be significant. The paired Wilcoxon test suggested by the framework is not used: runs of different algorithms are not paired, and with n = 5 it cannot reach $p < 0.05$.

Table 6. Mapping of the review framework's ten core metrics to this benchmark.

Review-framework metric	This benchmark	Status
Mission completion time (MCT)	mission_completion_time_s	measured
Task completion rate (TCR)	task_completion_rate_percent	measured
Total flight distance (TFD)	total_flight_distance_m	measured
Total energy consumed (TEC)	total_battery_consumed_percent	modelled as time (see 2.2)
Workload imbalance (WI)	workload_imbalance_tasks / _distance, Gini	measured (new)
Allocation time (AT)	alloc_first_assignment_s, alloc_all_assigned_s, ratio to MCT	measured (new)
Task reallocations (TR)	task_reallocations, per minute	measured (new)
Messages/bytes exchanged (MBE)	–	not instrumented
Minimum inter-drone distance (MIDD)	minimum, 1st/5th percentile, violations of 1.0 m	measured (extended)
Failure recovery time (FRT)	t_detect, t_reassign, T_recovery, rejoin, resume, transient safety	measured
– (added)	obstacle clearance, near misses, contacts; idleness; decision time	measured

5 Results

Figures show, for every algorithm, the mean (large dot), its 95 % confidence interval (line) and the individual runs (light dots), one column per scenario. Appendix A lists every value; Appendix C the significance tests. NMOPSO was run twice per scenario, the other algorithms five times: its means are measured, but its 95 % intervals (t with one degree of freedom) are wide, and with two runs no pairwise test involving NMOPSO can reach significance (the smallest attainable Mann–Whitney p for 2 against 5 runs is 0.095). NMOPSO is therefore compared by its values and mechanisms, and claims about it are worded as observations, not as significant differences.

5.1 Mission completion and reliability

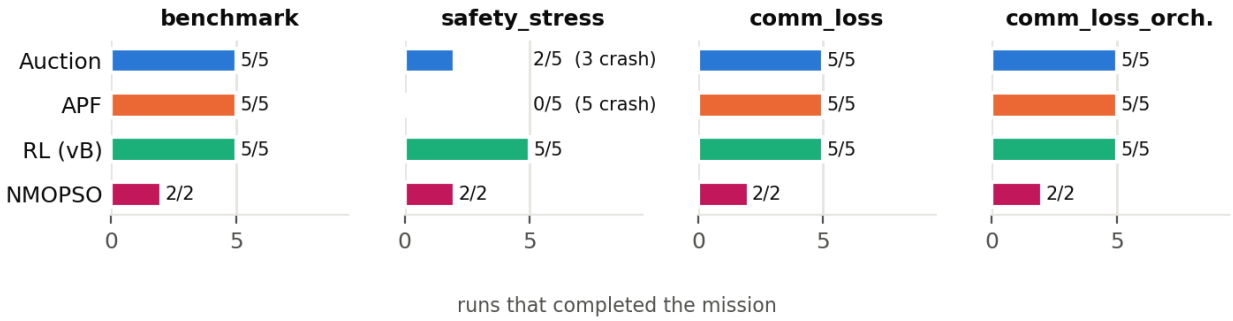


Figure 10. Runs that completed the mission, per scenario; runs in which a drone crashed are marked.

All four algorithms completed every nominal and communication-loss run, so the distinguishing scenario is safety_stress. There the RL planner completed 5 of 5 runs, NMOPSO 2 of 2, the auction 2 of 5 and APF 0 of 5; the failures are crashes (3 auction runs and 5 APF runs had a drone that hit an obstacle or another drone and was lost). Task completion rate in safety_stress is 90.0 ± 13.0 % for the auction, 87.5 ± 0.0 % for APF and 100 % for both planners.

Why. This scenario places three inspection points in a line along a rack aisle under the transit routes, one directly beside the centre corridor and one low inside an aisle. APF pulls each drone straight at its goal; near a rack the weak repulsion (it only balances the attraction at

about 0.6 m) loses to the attraction, and a drone threading between racks touches them or is trapped in a local minimum. The auction does not look at obstacles at all: it relies on its altitude layers and the corridor, and the task beside the corridor and the shared corridor itself bring drones of different layers together during descents, where the proximity hold does not always resolve the encounter. RL and NMOPSO plan every leg around the obstacle model before flying it, so a task inside an aisle is approached along a feasible path instead of being pulled at directly.

5.2 Efficiency: time, distance, idleness

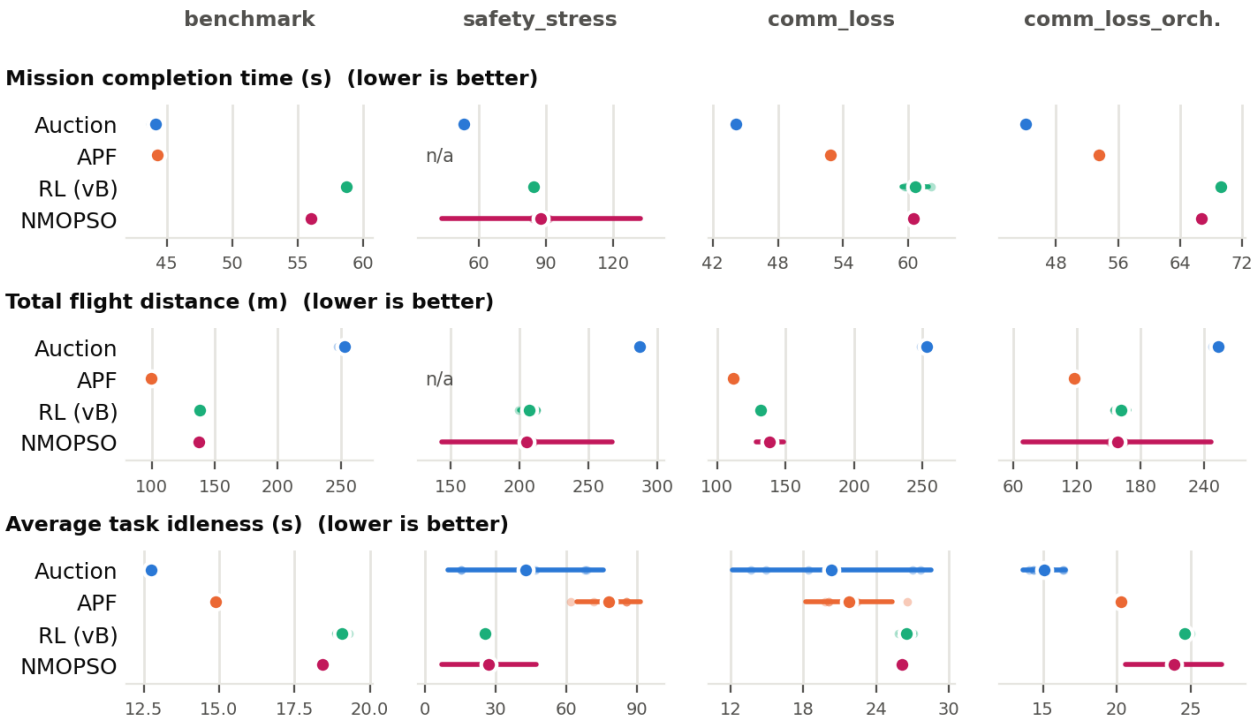


Figure 11. Efficiency metrics. Time and distance are averaged over completed runs only (APF has none in safety_stress).

Mission completion time. In the nominal benchmark the auction (44.1 ± 0.3 s) and APF (44.3 ± 0.2 s) are statistically tied and about 15 s faster than RL (58.7 ± 0.2 s). The tie hides two different paths to the same time: APF flies the shortest routes, while the auction flies 2.5 times as far (253.0 ± 3.8 m against 99.4 ± 0.4 m) but at transit altitude, above every obstacle, where nothing slows it down. RL is slower for two reasons: it detours around the obstacle model (flight distance 138.2 ± 3.1 m, 39 % above APF), and its planning stalls the coordinator while route decomposition runs (Section 5.5). NMOPSO isolates the second reason: its paths are planned around the same obstacle model and are as long as RL's (137.1 ± 3.4 m), but they come from the plan cache, so nothing stalls, and it finishes in 56.1 ± 0.3 s, about 2.7 s faster than RL. The remaining gap to the auction and APF is the detour itself.

Under communication loss the auction's mission time is unchanged (44.1 ± 0.3 s), because the drone without radio keeps inspecting its own tasks, while APF (52.8 ± 0.2 s) and RL (60.6 ± 1.2 s) lose the hovering drone's work until others take it over; NMOPSO (60.5 ± 0.3 s) behaves like RL. Losing the orchestrator costs RL most (69.2 ± 0.2 s, against 66.7 ± 0.5 s for NMOPSO).

Idleness. Average task idleness, which measures how fresh the inspection data is, favours the auction in every scenario without crashes (12.7 ± 0.0 s nominal against 14.9 ± 0.1 s for APF and 19.1 ± 0.2 s for RL and 18.4 ± 0.0 s for NMOPSO): each drone loops its tasks without waiting for ground-station decisions. In safety_stress the two planners have the lowest idleness

(25.5 ± 1.0 s for RL, 27.2 ± 20.2 s for NMOPSO), because the others leave tasks unvisited after a crash. Battery consumption follows mission time (the simulated battery is a time counter) and adds no separate information.

5.3 Safety: obstacles and separation

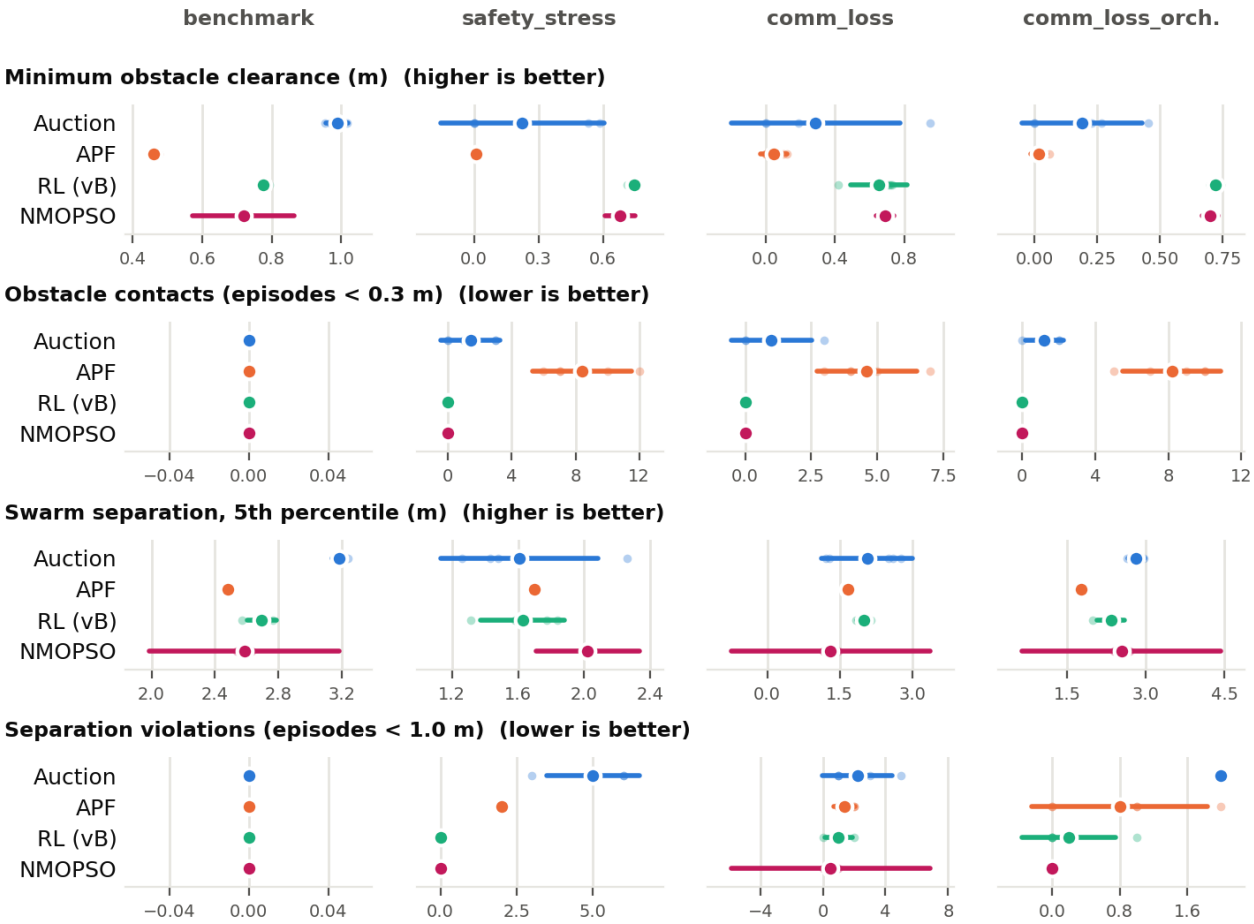


Figure 12. Safety metrics. Clearance is measured to the obstacle boxes for every algorithm; separation is the minimum over drone pairs at each instant.

Obstacle clearance. RL keeps the most consistent clearance: 0.78 ± 0.02 m nominal and 0.74 ± 0.02 m under stress, with zero contacts in every scenario. NMOPSO, which plans against the same obstacle model, is close behind (0.72 ± 0.15 m nominal, 0.68 ± 0.07 m under stress) and also never came within 0.6 m of an obstacle. The auction has the largest clearance in the nominal case (0.99 ± 0.03 m: its transits are above every obstacle) but collapses under stress (0.22 ± 0.38 m, 1.4 ± 1.9 contacts per run). APF flies closest to obstacles in every scenario (0.46 ± 0.01 m nominal, 8.4 ± 3.1 contacts per run under stress, and contacts even in the communication-loss scenarios, where reassigned tasks send drones on longer routes past racks).

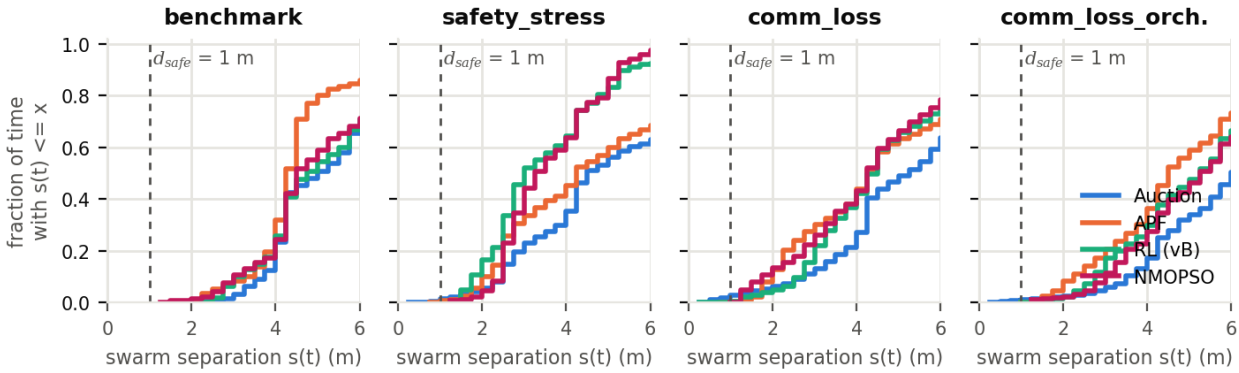


Figure 13. Distribution of the swarm separation s(t) over all runs (pooled): the fraction of time the closest pair is within x metres. Curves further right are safer; the dashed line is the 1.0 m safety radius.

Separation between drones. The percentiles and the distribution (Figure above) show more than the single minimum. In the nominal benchmark every algorithm keeps the drones well apart (no pair within 1.0 m); the auction has the widest margin (5th percentile 3.18 ± 0.04 m against 2.69 ± 0.09 m for RL and 2.48 ± 0.02 m for APF), because its transit altitudes are separated vertically by layer. Under stress the order reverses at the low end: RL is the only algorithm without a separation violation (0 per run, against 5.0 ± 1.5 for the auction and 2.0 ± 0.0 for APF), and its 1st percentile is the highest (1.51 ± 0.21 m). The auction's drones descend from their layers into the same aisle and corridor, where the proximity hold does not always resolve the encounter. APF's drone repulsion (3 m radius) prevents drone contacts under stress (0.0 ± 0.0 per run) but cannot protect against obstacles at the same time. NMOPSO also had no violation under stress, but its separation is the least consistent of the planners: each drone's path is optimised alone, without an inter-drone term, so the separation depends on which cached paths happen to cross (nominal minimum 1.59 ± 4.75 m over its two runs, against 2.36 ± 0.14 m for RL).

5.4 Fairness: workload balance

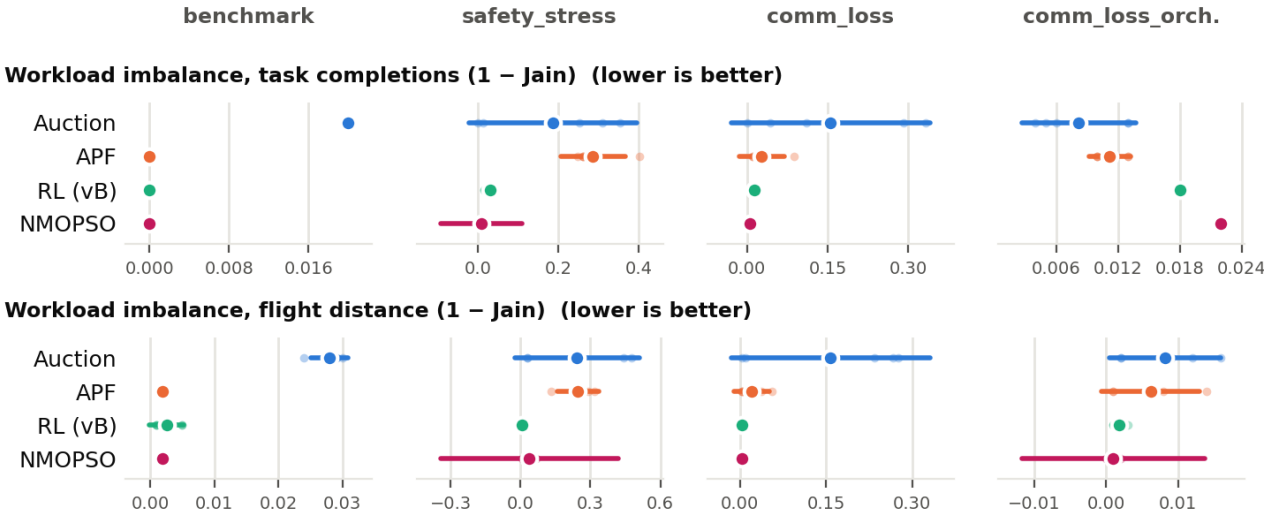


Figure 14. Workload imbalance on two workload definitions (0 = perfectly equal; at most 0.67 with three drones).

The shared auction caps every drone at ⌈M/N⌉ tasks, so the allocation itself is balanced in every algorithm, and nominal imbalance in task completions is near zero (0.020 ± 0.000, 0.000 ± 0.000, 0.000 ± 0.000, 0.000 ± 0.000 for auction, APF, RL and NMOPSO; the auction's small value is one extra visit by drone 0, which loops back to its first task before the cycle ends). Distance is less even for the auction (0.028 ± 0.003, against 0.002 ± 0.000 for APF and 0.003

± 0.003 for RL and 0.002 ± 0.000 for NMOPSO): its drone 0 flies about 100 m and its drone 2 about 65 m in one cycle, while APF, RL and NMOPSO drones fly within a few metres of each other. Faults are where imbalance becomes large. After a crash the surviving drones do the lost drone's work (safety_stress: 0.24 ± 0.27 for the auction and 0.25 ± 0.09 for APF, against 0.006 ± 0.006 for RL, which did not crash). Under communication loss the auction is again the least balanced (0.16 ± 0.19 on tasks, against 0.013 ± 0.008 for RL): the lost drone keeps working its own loop while its tasks are also given to another drone, so visits pile up on some drones.

5.5 Coordination overhead: allocation, reallocations, decision time

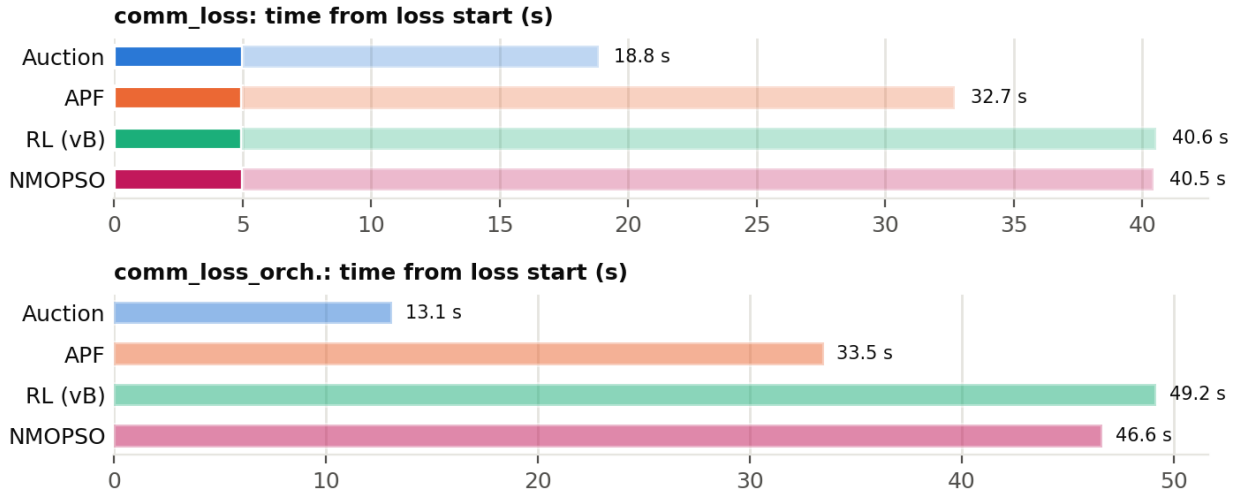


Figure 15. Coordination overhead. Decision time is measured in the coordinator, so the auction (on-board navigation) has none.

Allocation latency is the same for all four algorithms (5.16 ± 0.03 s nominal) because they share the auction, whose 5 s wait for peer states dominates it; the on-board auction step itself takes microseconds. The allocation-to-mission ratio (0.117 ± 0.001 for the auction, 0.087 ± 0.001 for RL) is well below the framework's 0.3 threshold for coordination-bound algorithms. **Task reallocations** are zero without faults; they appear only when the failure detector revokes a lost drone's tasks (communication loss of drone 1) and are absent when the orchestrator itself is lost, because nobody detects that loss.

Decision time is where APF and RL differ most: an APF decision takes 0.18 ± 0.02 ms on average and never more than about 1 ms, while an RL decision averages 9.6 ± 1.2 ms with worst cases of 1600 ± 93 ms. The worst cases are legs whose straight line is blocked, where the planner rolls out 193 candidates per sub-leg along an A* route. Because the planner runs on the coordinator's single thread, such a stall delays the commands of every drone, which is one of the reasons RL is slower. NMOPSO shows that this is an engineering choice, not a property of planning: its search is about fifty times slower than an RL plan (about 5 s per leg), yet its worst decision takes 1.9 ± 2.3 ms, because its legs are planned before the run and any other leg would be planned in a worker thread. In these scenarios every non-trivial leg came from the cache; the only legs planned during a run were 3–5 cm hops to the task a drone was already at.

5.6 Robustness: communication loss



Solid: detection (5 s heartbeat timeout); medium: until reassignment; light: until every task of the lost drone is visited again. Tick: lost drone back in service after the link returns.

Figure 16. Recovery after the 15 s radio blackout (means): detection, reassignment and the time until every task of the lost drone has been inspected again.



Figure 17. Recovery metrics in the two blackout scenarios.

Detection and reassignment take the heartbeat timeout, 5.00 ± 0.00 s, for every algorithm: detection and reallocation are platform functions shared by all four, and the reassignment follows detection within milliseconds. **Recovery differs because of what the lost drone does without radio.** An auction drone keeps flying its own task loop, so its tasks keep being inspected: coverage is restored after 18.8 ± 0.1 s, and after the link returns the drone is back in coordinated service within 3.8 ± 0.1 s. APF, RL and NMOPSO drones receive no movement commands and hold their last set-point; their tasks are covered only once the other drones have flown there (coverage restored 32.7 ± 0.2 s for APF, 40.6 ± 1.2 s for RL and 40.5 ± 0.6 s for NMOPSO), and after reconnection the hovering drone must win a task in a new auction and fly to it (26.5 ± 0.3 s, 25.6 ± 1.2 s and 25.4 ± 0.6 s).

Safety during and after the blackout. No algorithm lost separation during the blackout: the transient safety degradation index is zero in every run (time below 1.0 m from loss start to 10 s after the link returns: 0.00 ± 0.00 s, 0.00 ± 0.00 s, 0.00 ± 0.00 s and 0.00 ± 0.00 s). The close encounters recorded in these scenarios happen later, in the 150 s observation window that follows the first patrol cycle. The auction has the most (1.8 ± 1.4 contacts per run, against 0.6 ± 0.7 for APF, 0.2 ± 0.6 for RL and 0.0 ± 0.0 for NMOPSO), and they repeat at the same moments in every run: about 20 s after the link returns, between the reconnected drone 1 and drone 2, which had taken over one of its tasks, and later in the second patrol lap, between drones 0 and 1 (also in the orchestrator scenario). These are deterministic conflicts of the layered navigation, not random events; the nominal benchmark does not observe them only because it ends after the first cycle.

Losing the orchestrator (drone 0) exposes the single point of failure of the shared allocation layer: the loss is never detected and the lost drone's tasks are never reassigned in any algorithm; every run completes only because the link returns after 15 s. The auction is again least affected (coverage restored within the outage, 13.1 ± 0.2 s), while APF (33.5 ± 0.1 s), NMOPSO (46.6 ± 0.6 s) and RL (49.2 ± 0.2 s) wait for the link and then resume. RL takes longest because, on reconnection, it plans drone 0's remaining route while the stall delays all drones; NMOPSO, flying the same kind of paths from its cache, recovers about 2.6 s sooner.

5.7 Summary per metric

Table 7. Which algorithm is best at what (best = non-overlapping 95 % intervals or a consistent pattern across scenarios; see Appendix A and C).

Metric family	Best	Runner-up	Mechanism
Speed (nominal MCT, idleness)	Auction ≈ APF	NMOPSO, then RL	unobstructed transit layers / direct legs vs. detours; RL adds planning stalls
Path length	APF	NMOPSO ≈ RL	straight attraction vs. obstacle detours; auction climbs and uses the corridor
Task completion under stress	RL, NMOPSO	Auction	planned, obstacle-aware legs; APF and auction crash
Obstacle safety	RL	NMOPSO	planning around the obstacle model; APF repulsion too weak
Drone-to-drone safety	RL (stress, faults)	NMOPSO, APF	planned paths and yielding; NMOPSO optimises each drone alone; auction drones meet in aisles and corridor
Workload balance	tie (shared auction)	–	load cap of the auction; faults shift work
Allocation latency, reallocations	tie (shared auction)	–	same allocation layer
Decision time	APF	NMOPSO	closed-form forces; cached plans vs. RL's 193 rollouts per leg in the loop
Recovery from radio loss	Auction	APF	autonomous patrol without radio vs. hovering
Safety during radio loss	tie	–	no separation lost during any blackout; auction conflicts after reconnection

6 Discussion: which algorithm for a factory?

A factory imposes a lexicographic order on these metrics. A collision with a rack, a machine or a person is not a cost to be traded against seconds of mission time; it ends the mission and may injure someone. Coverage and inspection freshness come next, and only then path length or energy. Read in this order, the results point clearly in one direction while also showing what each approach would need.

Auction with layered navigation

The auction is the most efficient and most robust to radio loss in open layouts: its drones need no ground-station decisions, it has the lowest idleness, and a drone that loses its radio simply keeps patrolling. Its navigation, however, has no notion of obstacles; it is safe only while every obstacle is below the lowest transit layer and the corridor is free. In the stress scenario, which models a realistic aisle layout with low inspection points, it crashed in a majority of runs. It is suitable for open halls with high ceilings and well-separated, elevated inspection points, and for missions where radio dropouts are frequent, but not for dense racking.

APF

APF gives the shortest paths and negligible computation, and its drone-drone repulsion works well. Its obstacle avoidance is a local balance of forces and cannot guarantee clearance where inspection points sit next to obstacles: increasing the repulsion pushes drones away from the very points they must inspect, and local minima remain. It crashed in every stress run, came within 0.6 m of obstacles in the nominal scenario and touched them in both communication-loss scenarios. On its own it is not suitable for a cluttered factory; it is a good reactive safety layer underneath a planner, or a navigator for open areas.

RL leg planner

The RL planner and NMOPSO are the only approaches that completed every run of every scenario without touching an obstacle; RL has the most consistent clearance and no separation violation under stress. Its drones are not contact-free in every case: in the longer communication-loss runs one RL run had a drone contact in the second patrol lap, against several per run for the auction. It pays roughly a third more mission time and a larger worst-case decision time, and, like APF, its drones hover without radio. These costs are largely engineering, not intrinsic: planning in a worker thread removes the stalls, caching plans for the fixed patrol legs removes most planning altogether, and an on-board fallback (continue the last planned path, or return along it) would give it the auction's behaviour under radio loss.

NMOPSO

NMOPSO completed every run of every scenario without an obstacle or drone contact and with clearance close to RL's, so on this evidence it is as safe as RL in the cluttered layout. It was faster than RL in the nominal and orchestrator-loss scenarios and level with it under the loss of drone 1, because its legs are planned before the mission, which removes the planning stalls; under stress its two runs were slightly slower on average, within their wide interval. It has the lowest decision time of the two planners. The price is planning time: about 5 s per leg even vectorised (369 s for the authors' MATLAB code), so it fits a patrol whose legs are known in advance; a route that changes during the mission waits for a background plan. Its separation between drones is less consistent than RL's, because each drone's path is optimised alone. With two runs per scenario these are observations, not significant differences; five runs would put it on the same statistical footing as the others.

The allocation layer

All four share one weakness that the orchestrator scenario exposed: allocation and failure detection run on drone 0 only. A factory deployment should replicate them (for example, run the auctioneer at the ground station with a standby on a drone, or use a consensus-based auction) so that the loss of any single drone or link is detected and its tasks reallocated.

Recommendation. For an indoor factory with racks and machines, use a planning-based navigator (the RL planner, or NMOPSO with precomputed legs for a fixed patrol) with the market-based allocation, add an on-board continuation behaviour for radio loss, and replicate the allocator. Between the two planners, NMOPSO suits fixed patrol rounds in a static layout, where every leg can be planned ahead; RL suits layouts or routes that change, because it plans any leg in a fraction of a second. Use APF only as a local reactive layer, and layered auction navigation only in open, high-ceiling areas.

7 Conclusion

This benchmark compared a market-based auction, reactive potential fields, a PPO-trained leg planner and the NMOPSO particle swarm planner on one PX4/Gazebo platform, with shared allocation and instrumentation, across a nominal patrol, a spatial-conflict stress test and two communication-blackout faults, using the review framework's metrics extended with obstacle clearance, idleness, decision time and transient safety.

- No algorithm is best on every metric. Efficiency (time, path length, idleness) favours the auction and APF; safety and reliability favour the two planners, RL and NMOPSO; recovery from radio loss favours the auction.
- The decisive differences come from mechanisms, not tuning: whether navigation plans around obstacles (RL, NMOPSO) or reacts locally (APF) or ignores them (auction layers), and whether a drone can work without its radio (auction) or depends on the ground station (APF, RL, NMOPSO).
- In the cluttered stress scenario, both planners completed every run without contact, while APF crashed in every run and the auction in most. For factory deployment, where collisions are unacceptable, an obstacle-aware planner is required.
- Allocation latency, reallocation and failure detection are properties of the shared allocation layer; its single point of failure on drone 0 must be removed in any deployment.
- NMOPSO, run twice per scenario, matched RL's safety and was faster in the nominal and orchestrator-loss scenarios, because its legs are planned before the mission and never stall the control loop. RL's slower missions are therefore partly an engineering limit that asynchronous planning and plan caching remove.
- The remaining costs of the planners (hovering without radio, and for NMOPSO seconds per new leg) need an on-board continuation behaviour and, for changing routes, a fast planner such as RL.

The platform, the metrics and the analysis scripts are reusable: a new coordination algorithm implements one interface and is scored by the same suite, so the comparison can be extended to consensus auctions, bio-inspired allocation, larger swarms and hardware.

8 Limitations and threats to validity

- **Simulation only.** PX4 SITL in Gazebo, without wind, sensor noise beyond PX4 defaults or radio models other than the injected blackout. Absolute times and distances will differ on hardware; the ranking of mechanisms (Section 6) is the transferable result.
- **Five runs per cell, two for NMOPSO.** Outcomes that are bimodal (crash or no crash) produce wide intervals; ties in such cells mean "not separable with five runs". NMOPSO was run twice per scenario for time reasons: its means are measured, but its intervals are wide and its differences to the others cannot be tested for significance.
- **Different windows.** The benchmark and safety_stress runs end after the first patrol cycle, while the communication-loss runs are observed for 150 s, i.e. two to three cycles. Safety counts (contacts, violations) in the communication-loss scenarios therefore include later laps that the nominal benchmark never observes; they are not all caused by the blackout.
- **Blackout timing and NMOPSO's plan cache.** In both communication-loss scenarios the blackout starts 20 s into the mission, which in every run fell while the lost drone was inspecting a task. On reconnection its next leg therefore started at a task point and came from NMOPSO's plan cache instantly; the only legs planned during a run were 3–5 cm hops. A blackout that starts mid-flight, followed by a change of task, would leave the drone at a point no cached leg starts from: it would hover while the leg is planned in the background (about 5 s, about 20 s if the leg must be decomposed). NMOPSO's recovery times (Section 5.6) hold for these scenarios and would be longer for mid-flight blackouts, where RL plans a

leg in 0.1–1.6 s (but stalls every drone's commands while it does). The other drones are not affected: legs to reassigned tasks start at the task they just inspected and are cached.

- **One configuration per algorithm.** APF uses one parameter set, RL one policy (vB, seed 2) and NMOPSO the authors' published parameters. None was tuned for these scenarios; a parameter study could move APF along its speed–safety trade-off.
- **Shared allocation.** All four use the same auction, so allocation latency, reallocations and detection measure the platform, not the navigation method. Other allocation families (consensus auctions such as CBBA, bio-inspired allocation) are outside this benchmark.
- **Energy and messages.** The battery is a time counter, so no energy ranking is possible; message volume is not instrumented yet.
- **Fixable engineering effects.** RL's planning runs on the coordinator's single thread (worst decisions near 2 s), and allocation and failure detection run on drone 0 only. Both are platform choices that a deployment would change, not properties of the algorithms.
- **Three drones, one hall.** Scalability to larger swarms and other layouts is not tested.

9 Appendix

9.1 A. Results per scenario

Table 8. benchmark: mean ± 95 % CI half-width over the runs (time, distance, battery and the allocation ratio: completed runs only; NMOPSO: two runs).

Metric	Auction	APF	RL (vB)	NMOPSO
Runs completed / with crash	5/5 / 0	5/5 / 0	5/5 / 0	2/2 / 0
Task completion (%)	100 ± 0	100 ± 0	100 ± 0	100 ± 0
Mission time (s)	44.1 ± 0.3	44.3 ± 0.2	58.7 ± 0.2	56.1 ± 0.3
Flight distance (m)	253.0 ± 3.8	99.4 ± 0.4	138.2 ± 3.1	137.1 ± 3.4
Battery (%)	13.19 ± 0.07	13.24 ± 0.05	17.58 ± 0.06	16.77 ± 0.00
Average idleness (s)	12.7 ± 0.0	14.9 ± 0.1	19.1 ± 0.2	18.4 ± 0.0
Worst idleness (s)	44.1 ± 0.3	44.3 ± 0.2	58.7 ± 0.2	56.1 ± 0.3
Min obstacle clearance (m)	0.99 ± 0.03	0.46 ± 0.01	0.78 ± 0.02	0.72 ± 0.15
Obstacle near miss (drone-s)	0.00 ± 0.00	0.49 ± 0.11	0.00 ± 0.00	0.00 ± 0.00
Obstacle contacts	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
MIDD (m)	2.20 ± 0.06	1.99 ± 0.03	2.36 ± 0.14	1.59 ± 4.75
Separation p1 (m)	2.48 ± 0.10	2.02 ± 0.03	2.41 ± 0.12	1.71 ± 4.37
Separation p5 (m)	3.18 ± 0.04	2.48 ± 0.02	2.69 ± 0.09	2.58 ± 0.60
Violations < 1.0 m	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Drone contacts	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
WI tasks (1 − Jain)	0.020 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
WI distance (1 − Jain)	0.028 ± 0.003	0.002 ± 0.000	0.003 ± 0.003	0.002 ± 0.000
Gini distance	0.091 ± 0.005	0.023 ± 0.002	0.024 ± 0.017	0.024 ± 0.032
Task reallocations	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
All tasks assigned (s)	5.16 ± 0.03	5.16 ± 0.03	5.14 ± 0.05	5.16 ± 0.31
Allocation / MCT	0.117 ± 0.001	0.117 ± 0.001	0.087 ± 0.001	0.092 ± 0.000
Decision time mean (ms)	–	0.18 ± 0.02	9.59 ± 1.18	0.43 ± 0.25
Decision time max (ms)	–	0.5 ± 0.3	1599.6 ± 93.5	1.9 ± 2.3

Table 9. safety_stress: mean ± 95 % CI half-width over the runs (time, distance, battery and the allocation ratio: completed runs only; NMOPSO: two runs).

Metric	Auction	APF	RL (vB)	NMOPSO
Runs completed / with crash	2/5 / 3	0/5 / 5	5/5 / 0	2/2 / 0
Task completion (%)	90 ± 13	88 ± 0	100 ± 0	100 ± 0
Mission time (s)	53.2 ± 0.9	–	84.5 ± 0.7	87.6 ± 44.4
Flight distance (m)	287.2 ± 0.3	–	206.6 ± 6.7	205.3 ± 62.1
Battery (%)	15.93 ± 0.13	–	25.33 ± 0.23	26.25 ± 13.21
Average idleness (s)	42.9 ± 33.0	78.0 ± 13.4	25.5 ± 1.0	27.2 ± 20.2
Worst idleness (s)	201.1 ± 167.6	299.6 ± 1.4	84.5 ± 0.7	87.6 ± 44.4
Min obstacle clearance (m)	0.22 ± 0.38	0.01 ± 0.02	0.74 ± 0.02	0.68 ± 0.07
Obstacle near miss (drone-s)	1.83 ± 2.50	14.13 ± 4.59	0.00 ± 0.00	0.00 ± 0.00
Obstacle contacts	1.4 ± 1.9	8.4 ± 3.1	0.0 ± 0.0	0.0 ± 0.0
MIDD (m)	0.34 ± 0.16	0.76 ± 0.02	1.45 ± 0.21	1.13 ± 0.40
Separation p1 (m)	0.85 ± 0.32	1.12 ± 0.06	1.51 ± 0.21	1.39 ± 0.39
Separation p5 (m)	1.61 ± 0.48	1.70 ± 0.02	1.63 ± 0.25	2.02 ± 0.31
Violations < 1.0 m	5.0 ± 1.5	2.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Drone contacts	1.6 ± 1.1	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
WI tasks (1 − Jain)	0.186 ± 0.209	0.287 ± 0.080	0.030 ± 0.016	0.008 ± 0.102
WI distance (1 − Jain)	0.243 ± 0.266	0.246 ± 0.089	0.006 ± 0.006	0.037 ± 0.381
Gini distance	0.274 ± 0.223	0.293 ± 0.065	0.039 ± 0.023	0.092 ± 0.591
Task reallocations	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
All tasks assigned (s)	5.16 ± 0.02	5.16 ± 0.03	5.15 ± 0.04	5.15 ± 0.51
Allocation / MCT	0.097 ± 0.000	–	0.061 ± 0.001	0.059 ± 0.025
Decision time mean (ms)	–	0.21 ± 0.01	11.10 ± 0.64	0.48 ± 0.64
Decision time max (ms)	–	0.8 ± 0.4	1655.9 ± 157.6	1.6 ± 0.8

Table 10. comm_loss: mean ± 95 % CI half-width over the runs (time, distance, battery and the allocation ratio: completed runs only; NMOPSO: two runs).

Metric	Auction	APF	RL (vB)	NMOPSO
Runs completed / with crash	5/5 / 0	5/5 / 0	5/5 / 0	2/2 / 0
Task completion (%)	100 ± 0	100 ± 0	100 ± 0	100 ± 0
Mission time (s)	44.1 ± 0.3	52.8 ± 0.2	60.6 ± 1.2	60.5 ± 0.3
Flight distance (m)	252.9 ± 3.3	111.6 ± 0.5	131.6 ± 0.7	138.3 ± 10.1
Battery (%)	13.18 ± 0.09	15.79 ± 0.06	18.15 ± 0.36	18.11 ± 0.19
Average idleness (s)	20.3 ± 8.2	21.8 ± 3.6	26.5 ± 0.6	26.1 ± 0.5
Worst idleness (s)	83.6 ± 41.5	72.6 ± 26.7	73.9 ± 2.2	76.0 ± 4.4
Min obstacle clearance (m)	0.29 ± 0.49	0.05 ± 0.08	0.65 ± 0.16	0.69 ± 0.05
Obstacle near miss (drone-s)	0.49 ± 0.99	8.10 ± 2.01	0.08 ± 0.22	0.00 ± 0.00
Obstacle contacts	1.0 ± 1.5	4.6 ± 1.9	0.0 ± 0.0	0.0 ± 0.0
MIDD (m)	2.19 ± 0.03	1.65 ± 0.02	1.94 ± 0.14	1.28 ± 2.22
Separation p1 (m)	0.62 ± 0.36	1.29 ± 0.13	1.11 ± 0.06	1.07 ± 0.03
Separation p5 (m)	2.07 ± 0.94	1.67 ± 0.02	2.00 ± 0.15	1.31 ± 2.06
Violations < 1.0 m	2.2 ± 2.2	1.4 ± 0.7	1.0 ± 0.9	0.5 ± 6.4
Drone contacts	1.8 ± 1.4	0.6 ± 0.7	0.2 ± 0.6	0.0 ± 0.0
WI tasks (1 − Jain)	0.156 ± 0.185	0.027 ± 0.042	0.013 ± 0.008	0.006 ± 0.000
WI distance (1 − Jain)	0.158 ± 0.173	0.021 ± 0.031	0.004 ± 0.002	0.003 ± 0.000
Gini distance	0.192 ± 0.173	0.060 ± 0.055	0.032 ± 0.007	0.026 ± 0.006
Task reallocations	2.0 ± 0.0	2.0 ± 0.0	2.0 ± 0.0	2.0 ± 0.0
All tasks assigned (s)	5.17 ± 0.03	5.17 ± 0.02	5.16 ± 0.05	5.15 ± 0.37
Allocation / MCT	0.118 ± 0.001	0.098 ± 0.000	0.085 ± 0.002	0.086 ± 0.006
Decision time mean (ms)	–	0.20 ± 0.01	11.08 ± 2.10	0.50 ± 0.23
Decision time max (ms)	–	0.6 ± 0.3	1806.5 ± 310.8	2.0 ± 2.3
Detection (s)	5.00 ± 0.00	4.98 ± 0.05	4.98 ± 0.05	5.00 ± 0.01
Reassignment (s)	5.00 ± 0.00	4.98 ± 0.05	4.98 ± 0.05	5.00 ± 0.01
Coverage restored (s)	18.8 ± 0.1	32.7 ± 0.2	40.6 ± 1.2	40.5 ± 0.6
Back in service (s)	3.8 ± 0.1	26.5 ± 0.3	25.6 ± 1.2	25.4 ± 0.6
Min sep. lost drone (m)	2.34 ± 0.15	2.14 ± 0.11	1.94 ± 0.14	1.28 ± 2.22
Safety depth (m)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Time below 1.0 m (s)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

Table 11. comm_loss_orchestrator: mean ± 95 % CI half-width over the runs (time, distance, battery and the allocation ratio: completed runs only; NMOPSO: two runs).

Metric	Auction	APF	RL (vB)	NMOPSO
Runs completed / with crash	5/5 / 0	5/5 / 0	5/5 / 0	2/2 / 0
Task completion (%)	100 ± 0	100 ± 0	100 ± 0	100 ± 0
Mission time (s)	44.1 ± 0.4	53.6 ± 0.1	69.2 ± 0.2	66.7 ± 0.5
Flight distance (m)	252.7 ± 4.0	117.2 ± 0.2	161.4 ± 7.2	157.8 ± 88.5
Battery (%)	13.19 ± 0.11	16.03 ± 0.03	20.73 ± 0.06	19.96 ± 0.19
Average idleness (s)	15.1 ± 1.4	20.3 ± 0.0	24.6 ± 0.4	23.9 ± 3.3
Worst idleness (s)	56.5 ± 1.8	76.3 ± 0.2	76.2 ± 0.5	75.5 ± 3.2
Min obstacle clearance (m)	0.19 ± 0.24	0.02 ± 0.03	0.72 ± 0.02	0.70 ± 0.03
Obstacle near miss (drone-s)	1.01 ± 1.12	13.99 ± 2.90	0.00 ± 0.00	0.00 ± 0.00
Obstacle contacts	1.2 ± 1.0	8.2 ± 2.7	0.0 ± 0.0	0.0 ± 0.0
MIDD (m)	2.21 ± 0.09	1.40 ± 0.07	2.39 ± 0.18	2.24 ± 2.17
Separation p1 (m)	0.78 ± 0.14	1.29 ± 0.13	1.68 ± 0.55	1.20 ± 0.22
Separation p5 (m)	2.81 ± 0.16	1.77 ± 0.02	2.33 ± 0.25	2.54 ± 1.89
Violations < 1.0 m	2.0 ± 0.0	0.8 ± 1.0	0.2 ± 0.6	0.0 ± 0.0
Drone contacts	2.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
WI tasks (1 − Jain)	0.008 ± 0.005	0.011 ± 0.002	0.018 ± 0.000	0.022 ± 0.000
WI distance (1 − Jain)	0.008 ± 0.008	0.006 ± 0.007	0.002 ± 0.001	0.001 ± 0.013
Gini distance	0.043 ± 0.024	0.038 ± 0.024	0.022 ± 0.009	0.015 ± 0.108
Task reallocations	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
All tasks assigned (s)	5.17 ± 0.02	5.17 ± 0.02	5.14 ± 0.03	5.15 ± 0.15
Allocation / MCT	0.117 ± 0.001	0.096 ± 0.001	0.074 ± 0.001	0.077 ± 0.000
Decision time mean (ms)	–	0.20 ± 0.04	10.54 ± 1.13	0.46 ± 0.44
Decision time max (ms)	–	0.7 ± 0.4	1678.7 ± 65.0	1.7 ± 5.9
Coverage restored (s)	13.1 ± 0.2	33.5 ± 0.1	49.2 ± 0.2	46.6 ± 0.6
Back in service (s)	6.9 ± 0.2	18.5 ± 0.1	34.2 ± 0.2	31.6 ± 0.6
Min sep. lost drone (m)	2.21 ± 0.09	4.29 ± 0.01	4.28 ± 0.02	4.28 ± 0.26
Safety depth (m)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Time below 1.0 m (s)	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

9.2 B. Reproduction

`./scripts/run_suite.sh <name> 5` runs the suite (resumable); `python3 scripts/aggregate_suite.py experiment_results/suites/<name>` writes the tables; `presentation/.venv/bin/python presentation/scripts/build_report.py experiment_results/suites/<name>` rebuilds this report. Metric definitions with every threshold are in docs/METRICS.md; the RL training log is rl-nmopso-factory-patrol/LOOP.md.

9.3 C. Significance tests

Kruskal–Wallis p-value across the four algorithms, and pairwise Mann–Whitney U tests (two-sided, Holm-corrected p) with the rank-biserial correlation r (r > 0: the first algorithm of the pair has larger values; |r| = 1: complete separation). Only metrics with variation are listed.

Table 12. Non-parametric tests per scenario and metric.

Scenario	Metric	KW p	Pairs: p (Holm), r
benchmark	Mission time	0.004	Auc–APF: 0.381, -0.60; Auc–RL: 0.048, -1.00; Auc–NM: 0.381, -1.00; APF–RL: 0.048, -1.00; APF–NM: 0.381, -1.00; RL–NM: 0.381, +1.00
benchmark	Flight distance	0.003	Auc–APF: 0.048, +1.00; Auc–RL: 0.048, +1.00; Auc–NM: 0.286, +1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.857, +0.20
benchmark	Average idleness	0.002	Auc–APF: 0.048, -1.00; Auc–RL: 0.048, -1.00; Auc–NM: 0.286, -1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.286, +1.00
benchmark	Obstacle clearance	0.002	Auc–APF: 0.048, +1.00; Auc–RL: 0.048, +1.00; Auc–NM: 0.286, +1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.286, +1.00
benchmark	Separation p5	0.002	Auc–APF: 0.048, +1.00; Auc–RL: 0.048, +1.00; Auc–NM: 0.286, +1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.286, +0.80
benchmark	WI distance	0.012	Auc–APF: 0.044, +1.00; Auc–RL: 0.053, +1.00; Auc–NM: 0.304, +1.00; APF–RL: 1.000, +0.20; APF–NM: 1.000, +0.00; RL–NM: 1.000, -0.20
safety_stress	Mission time	0.117	Auc–RL: 0.286, -1.00; Auc–NM: 0.667, -1.00; RL–NM: 0.857, -0.20
safety_stress	Flight distance	0.117	Auc–RL: 0.286, +1.00; Auc–NM: 0.667, +1.00; RL–NM: 0.857, +0.20
safety_stress	Average idleness	0.026	Auc–APF: 0.159, -0.84; Auc–RL: 1.000, +0.20; Auc–NM: 1.000, +0.20; APF–RL: 0.048, +1.00; APF–NM: 0.381, +1.00; RL–NM: 1.000, -0.60
safety_stress	Obstacle clearance	0.005	Auc–APF: 0.724, +0.16; Auc–RL: 0.067, -1.00; Auc–NM: 0.282, -1.00; APF–RL: 0.067, -1.00; APF–NM: 0.282, -1.00; RL–NM: 0.282, +1.00
safety_stress	Obstacle contacts	0.004	Auc–APF: 0.057, -1.00; Auc–RL: 0.283, +0.60; Auc–NM: 0.560, +0.60; APF–RL: 0.044, +1.00; APF–NM: 0.283, +1.00; RL–NM: 1.000, +0.00
safety_stress	Separation p5	0.160	Auc–APF: 0.603, -0.60; Auc–RL: 1.000, -0.28; Auc–NM: 1.000, -0.60; APF–RL: 1.000, +0.20; APF–NM: 0.571, -1.00; RL–NM: 0.571, -1.00
safety_stress	Violations	0.001	Auc–APF: 0.035, +1.00; Auc–RL: 0.035, +1.00; Auc–NM: 0.146, +1.00; APF–RL: 0.024, +1.00; APF–NM: 0.082, +1.00; RL–NM: 1.000, +0.00
safety_stress	WI distance	0.011	Auc–APF: 1.000, -0.04; Auc–RL: 0.072, +1.00; Auc–NM: 0.762, +0.60; APF–RL: 0.072, +1.00; APF–NM: 0.381, +1.00; RL–NM: 0.710, -0.70
comm_loss	Mission time	0.003	Auc–APF: 0.048, -1.00; Auc–RL: 0.048, -1.00; Auc–NM: 0.286, -1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.857, +0.20
comm_loss	Flight distance	0.002	Auc–APF: 0.048, +1.00; Auc–RL: 0.048, +1.00; Auc–NM: 0.286, +1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.286, -1.00
comm_loss	Average idleness	0.331	Auc–APF: 1.000, -0.20; Auc–RL: 1.000, -0.28; Auc–NM: 1.000, -0.20; APF–RL: 0.190, -0.84; APF–NM: 1.000, -0.60; RL–NM: 1.000, +0.40
comm_loss	Obstacle clearance	0.034	Auc–APF: 0.986, +0.36; Auc–RL: 0.570, -0.60; Auc–NM: 0.986, -0.60; APF–RL: 0.072, -1.00; APF–NM: 0.393, -1.00; RL–NM: 0.986, +0.40
comm_loss	Obstacle contacts	0.004	Auc–APF: 0.075, -0.96; Auc–RL: 0.283, +0.60; Auc–NM: 0.560, +0.60; APF–RL: 0.044, +1.00; APF–NM: 0.283, +1.00; RL–NM: 1.000, +0.00
comm_loss	Separation p5	0.095	Auc–APF: 1.000, +0.20; Auc–RL: 1.000, +0.20; Auc–NM: 1.000, +0.60; APF–RL: 0.048, -1.00; APF–NM: 0.476, +1.00; RL–NM: 0.476, +1.00

Scenario	Metric	KW p	Pairs: p (Holm), r
comm_loss	Violations	0.274	Auc–APF: 1.000, +0.16; Auc–RL: 1.000, +0.40; Auc–NM: 1.000, +0.70; APF–RL: 1.000, +0.32; APF–NM: 1.000, +0.70; RL–NM: 1.000, +0.40
comm_loss	WI distance	0.133	Auc–APF: 0.798, +0.56; Auc–RL: 0.538, +0.68; Auc–NM: 0.798, +0.80; APF–RL: 1.000, +0.04; APF–NM: 0.937, +0.40; RL–NM: 0.840, +0.60
comm_loss	Coverage restored	0.003	Auc–APF: 0.055, -1.00; Auc–RL: 0.055, -1.00; Auc–NM: 0.203, -1.00; APF–RL: 0.048, -1.00; APF–NM: 0.203, -1.00; RL–NM: 0.857, +0.20
comm_loss	Back in service	0.006	Auc–APF: 0.065, -1.00; Auc–RL: 0.065, -1.00; Auc–NM: 0.271, -1.00; APF–RL: 0.302, +0.60; APF–NM: 0.286, +1.00; RL–NM: 0.857, +0.20
comm_loss_orchestrator	Mission time	0.002	Auc–APF: 0.048, -1.00; Auc–RL: 0.048, -1.00; Auc–NM: 0.286, -1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.286, +1.00
comm_loss_orchestrator	Flight distance	0.002	Auc–APF: 0.048, +1.00; Auc–RL: 0.048, +1.00; Auc–NM: 0.286, +1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.381, +0.60
comm_loss_orchestrator	Average idleness	0.002	Auc–APF: 0.048, -1.00; Auc–RL: 0.048, -1.00; Auc–NM: 0.286, -1.00; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.286, +1.00
comm_loss_orchestrator	Obstacle clearance	0.006	Auc–APF: 0.389, +0.36; Auc–RL: 0.072, -1.00; Auc–NM: 0.315, -1.00; APF–RL: 0.072, -1.00; APF–NM: 0.315, -1.00; RL–NM: 0.381, +0.80
comm_loss_orchestrator	Obstacle contacts	0.003	Auc–APF: 0.057, -1.00; Auc–RL: 0.097, +0.80; Auc–NM: 0.303, +0.80; APF–RL: 0.044, +1.00; APF–NM: 0.228, +1.00; RL–NM: 1.000, +0.00
comm_loss_orchestrator	Separation p5	0.003	Auc–APF: 0.048, +1.00; Auc–RL: 0.048, +1.00; Auc–NM: 0.381, +0.80; APF–RL: 0.048, -1.00; APF–NM: 0.286, -1.00; RL–NM: 0.571, -0.40
comm_loss_orchestrator	Violations	0.009	Auc–APF: 0.121, +0.80; Auc–RL: 0.034, +1.00; Auc–NM: 0.121, +1.00; APF–RL: 0.696, +0.44; APF–NM: 0.696, +0.60; RL–NM: 0.752, +0.20
comm_loss_orchestrator	WI distance	0.167	Auc–APF: 1.000, +0.36; Auc–RL: 0.502, +0.68; Auc–NM: 0.798, +0.80; APF–RL: 1.000, +0.36; APF–NM: 1.000, +0.60; RL–NM: 1.000, +0.40
comm_loss_orchestrator	Coverage restored	0.002	Auc–APF: 0.065, -1.00; Auc–RL: 0.065, -1.00; Auc–NM: 0.212, -1.00; APF–RL: 0.065, -1.00; APF–NM: 0.212, -1.00; RL–NM: 0.212, +1.00
comm_loss_orchestrator	Back in service	0.002	Auc–APF: 0.065, -1.00; Auc–RL: 0.065, -1.00; Auc–NM: 0.212, -1.00; APF–RL: 0.065, -1.00; APF–NM: 0.212, -1.00; RL–NM: 0.212, +1.00